Brief

Design & Implement Database

1. **Based on a hypothetical scenario below**

EasyBank is a newly established bank with **multiple branches** across the country (EasyCountry) that needs to set up its database structure. Just like a general bank system, they need to have details of their **customers** with information such as name, location, age, salary, contact details. Each customer would have a bank **account**, which could be a checking, savings or even a loan type account. Depending on the type of account, they would have different interest rates. Keep in mind that EasyBank accepts all types of customers such as babies, students, business owners and retirees. They also want to create an **online banking feature** for all their customers with a login and password where they can look at their transactions and history.

You are required to set up this database and populate it with a few examples.

Once you have set it up, EasyBank wants you to demonstrate the following required queries:

1. We want to offer loans to our customers; can you obtain a list of people and their contacts to our sales team? We want people who meet the following requirements:
    a. 18 or older
    b. Making over a certain amount (you can decide)
2. Compare the number of accounts by branches
3. Average salary difference between people under 30 and people above 50 years old

As you are their newly hired tech savvy person, EasyBank also wants to be able to have one of the following future functionalities:

1. **Predict a person's savings**

    What data would you use? How would you predict? What kind of method or model would you use? Can you provide an example of a query?

2. **Additional security method for online login**

What kind of data would you use? How would you check? Any tools that may be outside of the database?

2. **Your choice of database design – hotel management system, ecommerce system, etc.**

You can demonstrate the following queries on this database (these should be useful to know for your end user):

1. Show data analysis / results
2. Select a section of data based on a condition
3. Anything else you can think of

Future features:

1. AI/Machine learning feature
2. Additional security feature

Instructions

1. **Report containing:**
    a. A diagram/schema that demonstrates your database design, examples include:
        i. Entity Relationship (ER) model
        ii. Star Schema
        iii. Network Model
    b. Suggest solutions for “future functionalities”
    c. Process of design
    d. Screenshots of code and output
    e. Future improvements / thoughts & reflections
2. **SQL code**

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   a. Sets up the database based on the diagram
   b. Output of query and data analysis

3. **Presentation**
   a. Introduction/Background/Purpose
   b. Design process
   c. Demo
   d. Conclusion
   e. Q&A session

   Note: Unlimited attempts! Best score recorded

Guidelines

- Reports have no max/min limit. Make sure your report is well structured with titles, consistent styling throughout and relevant citation

- Presentations have no max/min limit. General presentation skills such as voice, tone, content, engagement with audience is graded

Submission Guidelines

Naming conventions

Firstname_Lastname_Database

Submission structure

Project document that contains the following:

- Presentation materials (if any)